

Domain-Adaptive Selective State-Space Learning for Cross-Subject Motor-Imagery EEG Decoding

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Abstract

Cross-subject motor-imagery electroencephalography (MI-EEG) decoding is limited by low signal-to-noise ratios, scarce labeled data, and strong distribution shifts across subjects and sessions. We propose a domain-adaptive selective state-space framework for this setting. A multi-kernel temporal convolutional front-end with depthwise spatial filtering and channel-group attention extracts local spectral-spatial features. Five selective state-space blocks with alternating forward and backward scans then model long-range temporal context at a cost linear in sequence length, and a convolutional shortcut with a grouped dilated temporal convolutional network fuses local and global information. To reduce cross-subject shift, the framework combines leakage-safe per-subject Riemannian alignment, gradient-reversal adversarial learning, and multi-kernel maximum mean discrepancy at the temporal and trial levels. After training, an information-maximization stage adapts only normalization parameters using unlabeled target trials. Target labels are never used for training or model selection. Code will be released at <https://github.com/USERNAME/REPOSITORY>.

Keywords: motor imagery, electroencephalography, brain-computer interface, selective state-space model, Mamba, domain adaptation, test-time adaptation

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1. Introduction

Motor-imagery brain–computer interfaces (MI-BCIs) translate internally generated movement intentions into commands without requiring overt muscular activity. Their non-invasive electroencephalography (EEG) interface and millisecond temporal resolution make them attractive for neurorehabilitation, assistive communication, and human–machine control. Reliable decoding nevertheless remains difficult. MI-related sensorimotor rhythms, observed as event-related desynchronization and synchronization (ERD/ERS) [1], are weak, noisy, non-stationary, and distributed over both time and scalp location. More importantly for practical deployment, their statistical distributions vary markedly among subjects and between recording sessions. A model trained for one user can therefore lose accuracy on a new user, while collecting and labeling a separate calibration set for every user is slow and burdensome. Cross-subject decoding consequently requires two capabilities at once: learning discriminative spatial–temporal structure from limited EEG trials and remaining robust to a previously unseen subject distribution.

Early MI decoders addressed the first requirement through signal processing and shallow machine learning. Common spatial pattern (CSP) learns spatial filters that maximize the variance contrast between two classes and remains one of the most influential MI feature extractors [2]. Its success also exposes its limitations: the learned projection is sensitive to the available training trials, conventional CSP is naturally binary, and performance depends strongly on the selected frequency interval. Filter-bank CSP (FBCSP) improved this pipeline by applying CSP to multiple sub-bands and selecting discriminative band-specific features [3]. Although effective and interpretable, CSP-family pipelines separate filtering, feature selection, and classification into manually designed stages. They do not learn task-dependent temporal, spectral, and spatial representations jointly, and their subject-specific covariance structure transfers poorly when the user changes.

End-to-end convolutional neural networks (CNNs) reduced this dependence on handcrafted features. Deep and shallow ConvNets demonstrated that temporal and spatial EEG filters can be learned directly from minimally processed signals [4]. EEGNet subsequently used depthwise and separable convolutions to encode frequency-specific spatial filters with a small parameter budget [5]. These designs provide useful inductive biases for EEG: short temporal kernels capture local oscillatory patterns, depthwise spatial kernels combine electrodes, and pooling suppresses small temporal translations.

Multi-scale or filter-bank convolution further reflects the fact that MI information is not confined to a single temporal scale. However, a CNN represents distant time points only after stacking layers, enlarging kernels, or using dilation. Its local bias alone may therefore be insufficient to model the full evolution of event-related desynchronization and synchronization within a trial.

Recent work has coupled convolution with attention or temporal sequence modules to capture local and global information jointly. EEG Conformer places self-attention after a convolutional tokenizer so that local temporal-spatial features can interact globally [6]. ATCNet combines attention, sliding temporal windows, and a temporal convolutional network (TCN), with dilated convolutions refining the time course of MI features [7]. More recently, TCFormer integrated a multi-kernel CNN, grouped-query Transformer layers, and a grouped TCN head [8]. This progression establishes a useful architectural principle: EEG-specific convolution should preserve local spectral and spatial structure, whereas a sequence encoder should capture global context. Yet self-attention explicitly constructs pairwise token interactions, giving quadratic time and memory complexity in sequence length [9]. This can be unnecessarily expensive for high-resolution or long-window EEG, and its flexible global operator can be data-hungry in the small-sample regimes typical of BCI experiments.

Architecture alone does not resolve inter-subject distribution shift. Domain generalization (DG) learns invariant representations from multiple labeled source subjects and deploys one model on an unseen target without accessing its data. Recent MI studies have explored spatial-temporal Transformers with DG [10], multi-source marginal and conditional alignment [11], and invariant spectral representations with correlation alignment [12]. DG is appealing for strict plug-and-play operation, but it cannot use the actual target distribution and may miss subject-specific shifts that were not represented by the sources. Domain adaptation (DA), in contrast, exploits unlabeled target trials while preserving a label-free target protocol. These settings must not be conflated: a method that observes unlabeled target EEG during training or adaptation is unsupervised DA rather than zero-shot DG.

EEG domain adaptation has developed along complementary geometric, discrepancy-based, and adversarial directions. Riemannian transfer learning and Euclidean alignment reduce covariance mismatch before classification by re-centering or whitening trials in a common space [13, 14]. At the representation level, deep adaptation networks minimize a multi-kernel maximum mean

discrepancy (MK-MMD) between source and target features [15]. Domain-adversarial neural networks instead train a feature extractor to confuse a domain discriminator through gradient reversal [16]; conditional adversarial variants use class predictions to preserve discriminative structure during alignment and have also been applied to MI-EEG [17]. For cross-subject MI decoding, HADANet combines a hierarchical convolutional extractor and hybrid attention with joint adversarial and MK-MMD alignment [18], and prototype-based adaptation networks additionally exploit local class information in the target domain [19]. These approaches are valuable but not risk-free. Global marginal alignment can mix class modes, and forcing a poorly matched or low-confidence target toward the sources can cause negative transfer. Moreover, aligning only the final trial vector may overlook a mismatch in the temporal representation from which that vector was produced.

Test-time adaptation offers another opportunity after source training. Information-maximization methods encourage confident yet globally diverse target predictions without requiring labels [20], while TENT showed that adapting only normalization statistics and affine parameters through entropy minimization can be an efficient response to distribution shift [21]. For EEG, unrestricted target fine-tuning would be hazardous because target datasets are small and model predictions can initially be unreliable. A constrained update space and a diversity term are therefore important safeguards against overfitting and single-class collapse.

Structured state-space models such as S4 provide a promising alternative to attention for modeling long sequences efficiently [22]. Building on this family, Mamba makes state-transition quantities input-dependent, enabling a recurrent state to selectively retain or discard information while maintaining linear scaling with sequence length [23]. Such selective memory is well matched to EEG, where discriminative activity is sparse, temporally structured, and embedded in noise. Recent MI-EEG decoders have accordingly combined convolutional feature extractors with Mamba blocks [24, 25]. These studies focus primarily on decoder architecture, leaving open how a selective state-space encoder should be combined with unlabeled target adaptation under a leakage-controlled cross-subject protocol. Moreover, replacing an EEG decoder with a generic state-space stack would discard useful domain knowledge. A pure unidirectional recurrence also introduces a directional bias even though an offline MI trial is available in full, and a sequence module should not be expected to replace explicit electrode filtering or local temporal ex-

traction.

Motivated by these observations, this paper proposes a hybrid, target-aware framework for cross-subject MI-EEG decoding. The model first applies three parallel temporal convolutions to the same EEG trial to capture complementary temporal scales. Their features are concatenated and processed by depthwise spatial convolution, which learns multiple electrode projections for each temporal feature. Grouped pointwise and temporal convolutions retain scale structure, and channel-group attention recalibrates the resulting branches. This convolutional representation is tokenized and passed through five lightweight selective state-space blocks. Their scan directions alternate forward and backward, allowing successive layers to incorporate both temporal directions without duplicating a bidirectional block at every depth. A direct convolutional shortcut is concatenated with the state-space output so that global contextualization does not erase local EEG evidence. Finally, a grouped TCN with increasing dilation refines multi-scale temporal dynamics before classification.

The proposed framework couples this encoder with hierarchical unsupervised domain adaptation that extends the hybrid adversarial–MK-MMD alignment of HADANet [18]. First, per-subject covariance alignment is fitted only on data legitimately available for that subject, preventing test-session leakage. During training, labeled source trials and unlabeled target-adaptation trials share the same feature extractor. A gradient-reversal discriminator promotes domain confusion, MK-MMD aligns pooled trial representations, and a lighter temporal MK-MMD aligns the time-averaged TCN feature sequences before trial aggregation. After training, an information-maximization procedure revisits only the unlabeled target-adaptation split and updates only BatchNorm affine parameters. The held-out target test sessions remain unseen until final inference. This combination is intended to preserve source discriminability, exploit target structure, and limit destructive target-specific updates.

The main contributions of this work are summarized as follows:

- We develop an EEG-specific encoder that combines multi-kernel temporal filtering, depthwise electrode filtering, a local convolutional shortcut, five alternating-direction selective state-space blocks, and a grouped dilated TCN. The design preserves local spatial–spectral priors while modeling long-range temporal context without pairwise self-attention.
- We introduce hierarchical cross-subject alignment that combines

leakage-safe covariance normalization, adversarial domain confusion, final-representation MK-MMD, and temporal MK-MMD.

- We incorporate a constrained source-free information-maximization stage in which only normalization affine parameters are adapted from unlabeled target trials, while a diversity objective discourages prediction collapse.
- We formulate a reproducible leave-one-subject-out evaluation protocol that keeps target labels out of training, adaptation, checkpoint selection, and preprocessing-statistic estimation, thereby separating the proposed UDA setting clearly from both supervised target calibration and target-free DG.

The remainder of this paper describes the datasets, model and adaptation objectives, leakage-controlled experimental protocol, planned result reporting, ablation design, discussion, and conclusions.

2. Datasets

Three public MI-EEG datasets are used to examine whether the same decoder transfers across different numbers of electrodes, classes, subjects, and recording sessions. Dataset access and event extraction are implemented through the MOABB/Braindecode interface; this choice provides a common representation of subjects, sessions, runs, and event windows and supports reproducible benchmarking [26]. Table 1 summarizes the properties relevant to the present work.

Table 1: Summary of the MI-EEG datasets (EOG channels excluded).

Dataset	Subjects	Channels	Sessions	Classes
BCI IV-2a	9	22	2	left hand, right hand, feet, tongue
BCI IV-2b	9	3	5	left hand, right hand
Zhou2016	4	14	3	left hand, right hand, feet

2.1. BCI Competition IV-2a

BCI Competition IV dataset 2a contains recordings from nine participants. Each participant completed two sessions on different days, and each session contains 288 trials distributed equally over imagined left-hand, right-hand, both-feet, and tongue movements. Signals were acquired from 22 EEG electrodes and three EOG electrodes at 250 Hz; only the EEG electrodes are supplied to the classifier. The original recordings were hardware band-pass filtered between 0.5 and 100 Hz and notch filtered at 50 Hz [27]. The implementation uses the event-defined MI interval exposed by MOABB, rescales volts to microvolts, and does not apply an additional frequency filter. The first recording session is denoted session 1 and the second is denoted session 2 throughout this paper.

2.2. BCI Competition IV-2b

BCI Competition IV dataset 2b contains two-class left- versus right-hand MI recordings from nine participants. Three bipolar EEG derivations centered on C3, Cz, and C4 were sampled at 250 Hz, together with three EOG channels that are excluded from decoding. Each participant completed five sessions: the first three form the calibration portion and the final two form the original evaluation portion [27]. The present loader follows this natural chronological separation. The event-defined MI window is used after removing the final 0.5 s through the configured stop offset. This dataset tests whether the method remains useful when spatial coverage is reduced from 22 electrodes to only three.

2.3. Zhou2016

The Zhou2016 dataset contains four participants, each recorded in three sessions separated by days to months. A session consists of two runs of 75 trials, with 25 trials per class, giving 450 trials per participant. The three MI classes are left hand, right hand, and feet. EEG was recorded from 14 electrodes at 250 Hz. Each trial contains a 5-s visual arrow period during which the indicated imagery is performed, followed by a 4-s rest period [28]. We therefore extract exactly the first 5.0 s from cue onset rather than including the subsequent rest interval. The public data are loaded through the MOABB representation, whose consistent dataset API is useful for avoiding dataset-specific indexing errors [26].

3. Proposed Method

Figure 1 presents the complete learning and inference pipeline. The source and target streams share one encoder, while supervision and adaptation objectives branch only after representation extraction. The figure also distinguishes joint UDA training from the post-training information-maximization stage and from the isolated held-out test sessions.

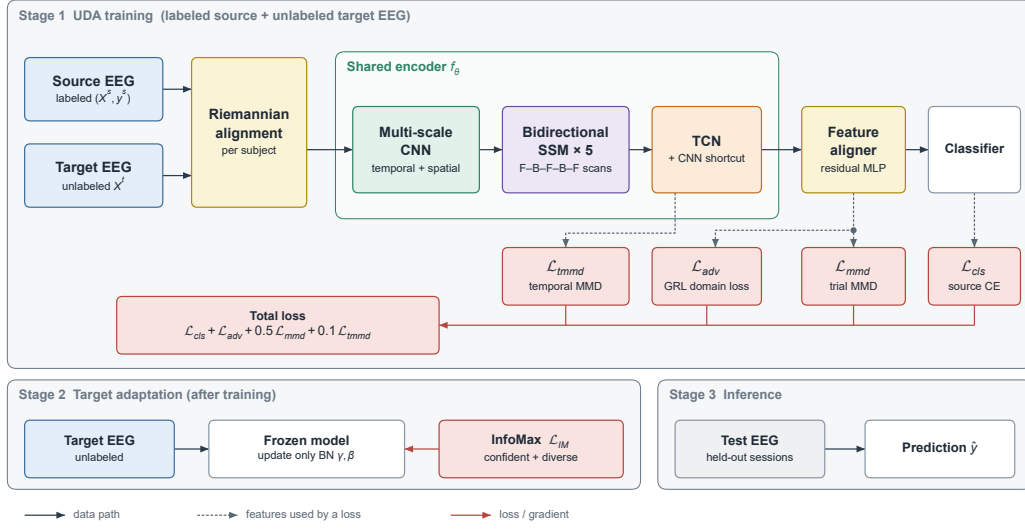


Figure 1: Overview of the proposed framework in one LOSO fold. Stage 1 trains the shared encoder with labeled source EEG and unlabeled target EEG; Stage 2 adapts only BatchNorm parameters on the unlabeled target sessions; Stage 3 evaluates the held-out target sessions.

3.1. Problem formulation

Let $\mathcal{S} = \{(X_i^s, y_i^s)\}_{i=1}^{N_s}$ denote labeled trials collected from all source subjects and let $\mathcal{T}_a = \{X_j^t\}_{j=1}^{N_t}$ denote the unlabeled adaptation-session trials of the held-out target subject. A trial is represented as $X \in \mathbb{R}^{C \times T}$, where C is the number of EEG channels and T is the number of samples. In each leave-one-subject-out (LOSO) fold, the model learns a classifier f_θ from $\mathcal{S}$ and may use the signal values, but never the labels, of $\mathcal{T}_a$. A disjoint set $\mathcal{T}_e$ containing later target sessions is reserved for a single final evaluation. No quantity estimated from $\mathcal{T}_e$ is used for normalization, covariance alignment, optimization, checkpoint selection, or hyperparameter selection.

3.2. Leakage-safe signal alignment

The implementation optionally applies a separate Riemannian whitening operator to every subject. For one subject, covariance matrices are estimated from that subject’s available adaptation or training trials with Oracle Approximating Shrinkage. Their affine-invariant Riemannian mean is denoted by $\bar{R}$. Every trial belonging to the same subject is transformed as

$$\tilde{X} = \bar{R}^{-1/2} X. \quad (1)$$

For a source subject, $\bar{R}$ is fitted using only that source subject’s training sessions and then applied to its validation sessions. For the target subject, it is fitted only on $\mathcal{T}_a$ and then frozen before application to $\mathcal{T}_e$. This construction follows the motivation of Riemannian transfer and alignment methods [13, 14], while preventing evaluation-session covariance from leaking into the fitted transform. Standardization statistics are likewise estimated without target-test labels or target-test signals.

3.3. Multi-scale temporal and spatial convolution

The convolutional front-end and its tensor dimensions are detailed in Fig. 2. Its output H_c feeds both the selective state-space encoder (Section 3.4) and a direct local shortcut; the two paths are reunited before the TCN (Section 3.5).

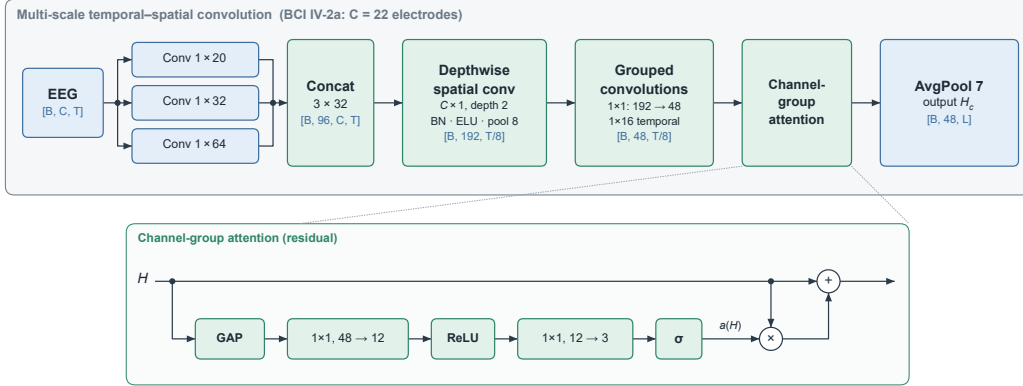


Figure 2: Multi-scale temporal-spatial convolutional front-end, illustrated for BCI IV-2a. Three temporal kernels are followed by depthwise electrode filtering, grouped channel reduction, grouped temporal refinement, and residual channel-group attention, producing the sequence H_c .

Following the multi-kernel temporal front-end of TCFormer [8], the convolutional block first reshapes a mini-batch to $B \times 1 \times C \times T$. Three temporal branches then process the same trial using kernels of length 20, 32, and 64 samples:

$$H_k = \text{BN}(\text{Conv}_{1 \times k}(X)), \quad k \in \{20, 32, 64\}. \quad (2)$$

Asymmetric zero padding for even kernels preserves the original time length. Each branch produces 32 maps, and concatenation gives 96 maps while retaining the identity of the three scales.

A depthwise convolution whose spatial kernel spans all C electrodes is applied independently to every temporal map. With a depth multiplier of two, each of the 96 inputs learns two electrode projections, resulting in 192 maps. Batch normalization and ELU activation follow the spatial filter. Average pooling by eight and dropout reduce temporal resolution and regularize the representation. A grouped 1×1 convolution then maps 192 channels to 48 channels, with 16 channels retained for each temporal-scale group. A grouped temporal convolution with kernel length 16 further refines each scale without mixing the groups.

Channel-group attention assigns one data-dependent sigmoid coefficient to each temporal-scale group. Global average pooling is followed by grouped 1×1 projections $48 \rightarrow 12 \rightarrow 3$, ReLU, and sigmoid. Its output is used residually:

$$H_{\text{att}} = H + H \odot a(H), \quad (3)$$

where the three coefficients in $a(H)$ are broadcast over the 16 channels in their corresponding groups. A second average-pooling layer with factor seven and dropout produces a sequence $H_c \in \mathbb{R}^{B \times 48 \times L}$.

3.4. Alternating selective state-space encoder

Before sequence modeling, a 1×1 convolution, batch normalization, and SiLU activation mix the 48 convolutional channels. The sequence is then processed by five lightweight selective state-space blocks in the order forward-backward-forward-backward-forward. A backward block reverses the temporal axis before its recurrence and reverses the output afterward. Consequently, the stack incorporates evidence from both temporal directions while each individual block retains the parameter cost of a single scan. The internal structure of one block is shown in Fig. 3.

Each block normalizes its input tokens and projects them into a value stream and a SiLU-gated stream. A causal depthwise temporal convolution

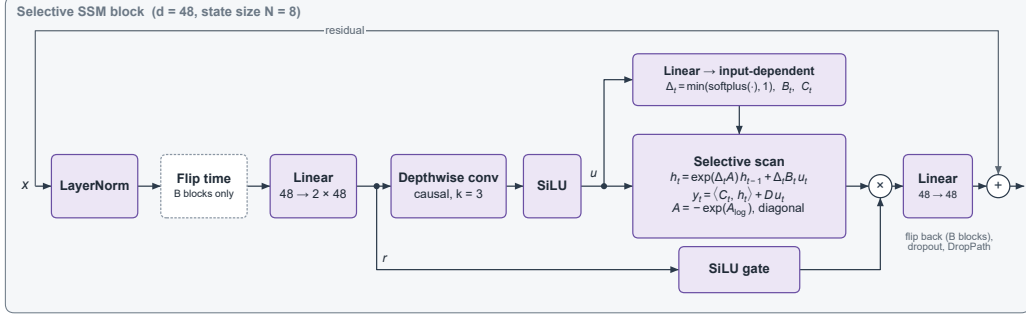


Figure 3: Structure of one selective SSM block. Backward (B) blocks reverse the time axis before the mixer and restore it afterwards; forward (F) blocks skip both flips.

of length three followed by SiLU supplies local context to the value stream, giving u_t . Following the diagonal state-space formulation used in structured SSMs [22], the input-dependent step size Δ_t and state projections B_t and C_t parameterize a selective recurrence of the form

$$A = -\exp(A_{\log}), \quad (4)$$

$$h_t = \exp(\Delta_t A) \odot h_{t-1} + \Delta_t B_t \odot u_t, \quad (5)$$

$$z_t = C_t^\top h_t + d_{\text{skip}} \odot u_t. \quad (6)$$

The block output is projected after modulation by the learned gate and added to the residual stream. The stable negative diagonal A and input-dependent transition quantities implement a lightweight selective-memory mechanism inspired by Mamba [23]. This is a simplified PyTorch implementation rather than the official Mamba block or its hardware-aware parallel selective-scan kernel. The code uses a state dimension of eight, dropout of 0.4, and a depth-dependent stochastic-depth schedule. Unlike pairwise self-attention, the recurrent scan has linear sequence-length scaling.

3.5. Local-global fusion, TCN, and classification

The output of the fifth state-space block is reduced from 48 to 16 channels with a pointwise convolution. It is concatenated with the untouched 48-channel convolutional sequence H_c , yielding 64 channels. This shortcut protects local temporal-spatial evidence from being overwritten by the global sequence encoder. The fusion, TCN, and classification head are shown in Fig. 4.

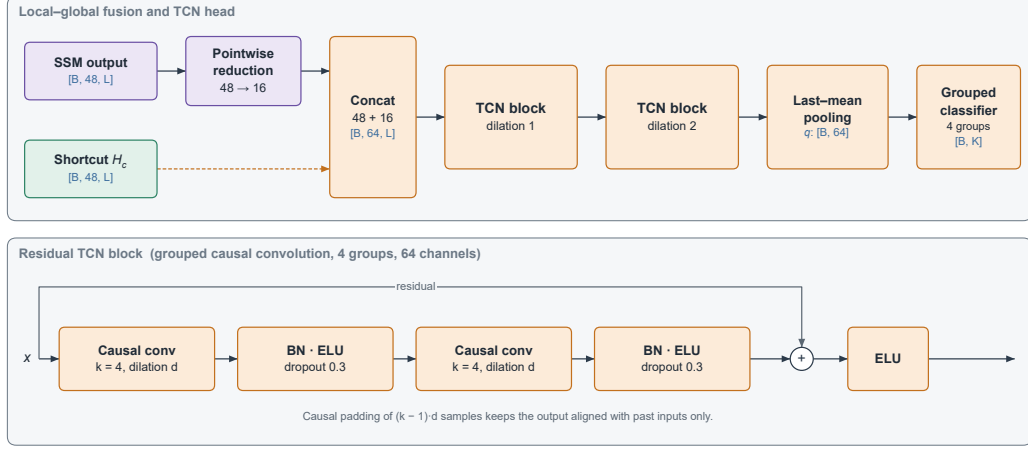


Figure 4: Local-global fusion and TCN head. Top: the reduced SSM output is concatenated with the shortcut H_c and refined by two residual TCN blocks before last-mean pooling and grouped classification. Bottom: structure of one residual TCN block.

Two residual TCN blocks then refine the fused sequence. Each block contains two causal grouped one-dimensional convolutions with kernel length four, batch normalization, ELU, and dropout. Dilations one and two enlarge the receptive field while preserving temporal order. The final trial representation is

$$q = \frac{1}{2} \left(H_{\text{TCN}}[:, :, -1] + \frac{1}{L} \sum_{t=1}^L H_{\text{TCN}}[:, :, t] \right), \quad (7)$$

which balances the last causal state with the mean over the whole sequence. A grouped norm-constrained 1×1 classifier produces one class-logit contribution per feature group, and learned softmax weights combine the four group contributions.

3.6. Hierarchical unsupervised domain adaptation

The 64-dimensional pooled representation first passes through a residual feature aligner comprising layer normalization and a $64 \rightarrow 128 \rightarrow 64$ GELU multilayer perceptron with dropout 0.3 and a final layer normalization. Its residual scale is initialized to 0.1 so that adaptation begins close to the source-trained representation.

As shown in Fig. 1, temporal discrepancy is measured before trial aggregation, whereas pooled-feature discrepancy, adversarial domain confusion,

and source classification act on the aligned trial vector. Only $\mathcal{L}_{\text{cls}}$ consumes class labels, and the later information-maximization stage updates only normalization affine parameters.

Three complementary losses are then computed. First, source cross-entropy preserves class discrimination:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N_s} \sum_i \log p_{\theta}(y_i^s | X_i^s). \quad (8)$$

Second, a gradient-reversal layer connects pooled source and target features to a binary domain discriminator [16]. The discriminator uses projections $64 \rightarrow 128 \rightarrow 64 \rightarrow 1$, LeakyReLU activations, and dropout 0.3. Its reversal coefficient follows

$$\alpha(p) = \frac{2}{1 + \exp(-10p)} - 1, \quad (9)$$

where p is normalized training progress. The resulting binary cross-entropy is denoted $\mathcal{L}_{\text{adv}}$.

Third, MK-MMD measures source–target discrepancy using five Gaussian kernels with data-dependent bandwidth [15]. Features are first normalized to unit ℓ_2 norm, and the implementation removes within-domain diagonal terms to obtain an unbiased estimate when both mini-batches contain at least two trials. MK-MMD is evaluated on both the aligned trial vectors, giving $\mathcal{L}_{\text{mmd}}$, and the time-averaged TCN feature sequences, giving $\mathcal{L}_{\text{tmmd}}$. The temporal term introduces no additional trainable parameters and discourages alignment that matches only the final pooled vector.

The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{adv}} + 0.5 \mathcal{L}_{\text{mmd}} + 0.1 \mathcal{L}_{\text{tmmd}}. \quad (10)$$

Only source labels contribute to Eq. (10); the target branch receives EEG samples without labels.

3.7. Constrained information-maximization adaptation

After source/UDA training, all model parameters are frozen except the affine weights and biases of BatchNorm layers. Five passes over $\mathcal{T}_a$ are performed with Adam at a learning rate of 10^{-4} . Let p_j denote a target prediction and $\bar{p} = N_t^{-1} \sum_j p_j$. The information-maximization objective is

$$\mathcal{L}_{\text{IM}} = \frac{1}{N_t} \sum_j H(p_j) - H(\bar{p}). \quad (11)$$

The first term promotes confident predictions, whereas the second encourages a diverse target class marginal and reduces the risk of collapse to one class [20, 21]. Importantly, the clean protocol applies this stage only to $\mathcal{T}_a$. The held-out test loader $\mathcal{T}_e$ is never passed to the adaptation function.

4. Experimental Protocol

4.1. Strict LOSO session protocol

One subject is held out in each LOSO fold. All remaining subjects are sources; the held-out participant is the target. Table 2 specifies the chronological session split. The target-adaptation sessions are available only as unlabeled EEG. Target-test sessions are hidden from preprocessing, training, adaptation, and checkpoint selection until one final evaluation. This is the primary protocol of the paper and is the only protocol to be used for headline comparisons.

Table 2: Session allocation in each LOSO fold.

Dataset	Source subjects (labeled)		Target subject	
	Training	Validation	Adaptation (unlabeled)	Test (held out)
BCI IV-2a	session 1	session 2	session 1	session 2
BCI IV-2b	sessions 1–3	sessions 4–5	sessions 1–3	sessions 4–5
Zhou2016	sessions 1–2	session 3	session 1	sessions 2–3

For BCI IV-2a, this split yields nine folds; for BCI IV-2b, nine folds; and for Zhou2016, four folds. The target identifier changes between folds, but the split rule and hyperparameters remain fixed. The checkpoint with the highest source-validation accuracy is selected. Neither target-adaptation predictions nor target-test metrics participate in checkpoint selection. The same target-adaptation loader is used for adversarial/MK-MMD training and the subsequent information-maximization stage.

4.2. Preprocessing and augmentation

Only EEG channels are retained. Signals are resampled to 250 Hz and converted from volts to microvolts. No additional digital band-pass filter

is applied in the model configuration, because the public BCI Competition recordings already include acquisition filtering and the network learns its temporal filters from the signal. BCI IV-2a uses the complete MOABB event window; BCI IV-2b removes 0.5 s from the end of the event window; Zhou2016 uses exactly 5.0 s beginning at cue onset.

Riemannian whitening, when active, follows the session-isolated fitting rule in Section 3.2. Z-score statistics are fitted using the source-training partition and then frozen. Inter-trial segment reconstruction augmentation is applied only to labeled source mini-batches: segments are recombined among trials of the same source class, so it neither reads target labels nor creates mixed source–target examples. Zhou2016 is run without this augmentation to preserve its intended dataset-specific setting.

4.3. Training configuration

All experiments are executed in Kaggle notebooks using one NVIDIA Tesla T4 GPU with 16 GB memory. BCI IV-2a and BCI IV-2b use a mini-batch size of 48 and a maximum of 125 epochs. Zhou2016 uses a mini-batch size of 32 and 100 epochs. The BCI Competition runs use Adam with an initial learning rate of 9×10^{-4} , $\beta_1 = 0.5$, and weight decay 10^{-3} ; the Zhou2016 setting uses an initial learning rate of 10^{-3} and weight decay 10^{-4} . A linear warm-up over the first three LOSO epochs is followed by cosine decay. The random seed, data split, and initialization procedure are held fixed for direct architectural ablations. Final reporting should additionally repeat each complete LOSO experiment with multiple seeds to quantify optimization variance.

4.4. Evaluation measures and reporting

The primary measures are trial-level classification accuracy and Cohen’s κ , reported as mean and standard deviation across target-subject folds. Per-subject accuracies are reported for every dataset so that a mean is not interpreted without its cross-subject dispersion. Accuracy differences between the proposed method and each baseline are tested on paired subject folds with a two-sided Wilcoxon signed-rank test. Parameter count, single-trial inference latency, and throughput are secondary efficiency measures. Results from a protocol that exposes all target sessions are explicitly marked auxiliary and are never compared directly with the strict held-out-session results.

Inference efficiency is measured after training on the same NVIDIA Tesla T4 GPU used for training, with preprocessing and host-device transfer excluded. Every network is placed in evaluation mode and executed under inference-only semantics with the BCI IV-2a input shape (22 channels $\times$ 1000 samples). Single-trial latency is measured with batch size one and explicit CUDA synchronization; each measurement uses 100 warm-up iterations followed by 1000 timed iterations. Throughput is measured independently with a common batch size of 32 and is reported in trials per second; it is not inferred by taking the reciprocal of batch-one latency. Information-maximization adaptation is a one-time target-subject operation and is therefore timed separately in seconds per target subject rather than included in inference latency.

4.5. Baseline comparison policy

Tables 3–5 group target-free source-only methods separately from methods that consume unlabeled target-adaptation EEG (UDA). Every baseline must be trained and evaluated using exactly the subject folds and session allocation in Table 2: all non-target participants are source subjects, one complete participant is held out as the target, and only the designated earlier target sessions may be used without labels by UDA methods. Published headline values obtained with within-subject splits, random trial splits, different windows, or access to target-test EEG are not copied into the tables. Each baseline is rerun from its public implementation under the common preprocessing and protocol; otherwise its entry is marked not available rather than treated as directly comparable.

The source-only group comprises three deep decoders that combine convolution with attention or temporal convolution: EEG Conformer [6], ATCNet [7], and TCFormer [8]. These models receive no target EEG before final inference. The UDA group includes a gradient-reversal DANN baseline [16], an MK-MMD/DAN baseline [15], and conditional adversarial adaptation [17].

The proposed method is reported in three variants that share the same encoder and differ only in how much target EEG they use (Fig. 1):

- **Ours-S** (source-only): trained with $\mathcal{L}_{\text{cls}}$ on the labeled source subjects only. No target EEG is seen before inference, so this variant is directly comparable with the source-only baselines and isolates the contribution of the encoder.

- **Ours-U** (UDA): Stage 1 only, trained with the complete objective in Eq. (10) using the unlabeled target-adaptation EEG, without information-maximization adaptation.
- **Ours** (complete framework): Ours-U followed by the Stage 2 information-maximization adaptation of BatchNorm affine parameters (Section 3.7).

The difference between Ours-U and Ours-S therefore measures the gain from domain alignment, and the difference between Ours and Ours-U measures the gain from information-maximization adaptation.

5. Results

This section is intentionally reserved until all clean-protocol runs, repeated seeds, and statistical tests have been completed. No preliminary or all-session value is inserted into the primary tables. Each dataset is reported in its own table with an identical layout: target-test accuracy and κ for every held-out subject, their mean and standard deviation across subjects, and the Wilcoxon p -value of each method’s accuracy against the complete proposed framework.

5.1. BCI Competition IV-2a

Table 3 reports the four-class strict-session LOSO results for the nine BCI IV-2a target subjects.

5.2. BCI Competition IV-2b

Table 4 reports the binary left- versus right-hand results for the nine BCI IV-2b target subjects, whose recordings contain only three bipolar EEG derivations.

5.3. Zhou2016

Table 5 reports the three-class results for the four Zhou2016 target subjects. With only four folds, the smallest attainable two-sided Wilcoxon p -value is 0.125; the p -values for this dataset are therefore reported for completeness but cannot reach conventional significance.

5.4. Computational efficiency

Table 6 compares model size and inference cost on the Tesla T4 GPU.

Table 3: Per-subject LOSO accuracy (%) and κ on BCI IV-2a (four classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA				
		Conformer	ATCNet	TCFormer	Ours-S	DANN	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[16]	[15]	[17]		
S1	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S2	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S3	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S4	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S5	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S6	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S7	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S8	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S9	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
Mean	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
SD	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
p		—	—	—	—	—	—	—	—	—

Table 4: Per-subject LOSO accuracy (%) and κ on BCI IV-2b (two classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA				
		Conformer	ATCNet	TCFormer	Ours-S	DANN	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[16]	[15]	[17]		
S1	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S2	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S3	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S4	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S5	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S6	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S7	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S8	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S9	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
Mean	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
SD	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
p		—	—	—	—	—	—	—	—	—

Table 5: Per-subject LOSO accuracy (%) and κ on Zhou2016 (three classes). Ours-S: source-only; Ours-U: UDA without IM adaptation; Ours: UDA with IM adaptation (Section 4.5). p : Wilcoxon signed-rank test of accuracy against Ours.

Subj.		Source-only				UDA				
		Conformer	ATCNet	TCFormer	Ours-S	DANN	DAN	CDAN	Ours-U	Ours
		[6]	[7]	[8]		[16]	[15]	[17]		
S1	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S2	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S3	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
S4	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
Mean	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
SD	Acc.	—	—	—	—	—	—	—	—	—
	κ	—	—	—	—	—	—	—	—	—
p		—	—	—	—	—	—	—	—	—

Table 6: Computational efficiency on a Tesla T4 GPU (BCI IV-2a input, 22×1000). Ours-S, Ours-U, and Ours are defined in Section 4.5.

Method	Parameters (M)	Latency (ms/trial)	Throughput (trials/s)	IM adaptation (s/subject)
EEG Conformer [6]	—	—	—	N/A
ATCNet [7]	—	—	—	N/A
TCFormer [8]	—	—	—	N/A
Ours-S	—	—	—	N/A
Ours-U	—	—	—	N/A
Ours	—	—	—	—

6. Ablation Study

The ablation study uses an auxiliary all-session target view as requested: sessions 1–2 for BCI IV-2a, sessions 1–5 for BCI IV-2b, and sessions 1–3 for Zhou2016. In this analysis, all target EEG may be exposed without labels to the adaptation procedure and is subsequently scored with labels. This is a transductive diagnostic protocol, not the strict protocol in Table 2. It provides more target samples and lower-variance component comparisons, but it is easier and can be optimistic because the same session distribution is visible during adaptation and scoring. Consequently, its numbers must be confined to this section and must not be presented as the main benchmark.

The ablations isolate both architectural and adaptation components: removal of covariance alignment; removal of channel-group attention; replacement of the alternating scans by forward-only scans; removal of the grouped TCN; source-only training without adversarial or MMD losses; removal of temporal MMD; and removal of information-maximization adaptation. Only one component is changed at a time, with the same subject folds, seed set, training budget, and source-validation checkpoint rule. Table 7 reports both accuracy and κ for each dataset.

Table 7: Ablation results under the all-session protocol: mean accuracy (%) and κ .

Configuration	BCI IV-2a		BCI IV-2b		Zhou2016	
	Acc.	κ	Acc.	κ	Acc.	κ
Complete proposed framework	–	–	–	–	–	–
w/o Riemannian alignment	–	–	–	–	–	–
w/o channel-group attention	–	–	–	–	–	–
Forward-only state-space scans	–	–	–	–	–	–
w/o grouped TCN	–	–	–	–	–	–
w/o adversarial and MMD losses	–	–	–	–	–	–
w/o temporal MMD	–	–	–	–	–	–
w/o information-maximization	–	–	–	–	–	–

7. Discussion

The experimental design separates two questions that are often blended in cross-subject MI studies. The strict protocol asks whether early unlabeled recordings from a new user can improve decoding on later, unseen sessions. The all-session ablation asks which components are useful in a transductive setting where the complete target distribution is visible without labels. Reporting these settings separately is essential: target labels are absent in both, but access to future-session signals changes the information available to the algorithm and can materially change accuracy.

The proposed architecture is motivated by complementary inductive biases rather than by replacing every operation with a generic sequence model. Temporal kernels of different lengths capture local MI patterns; depthwise spatial convolution specializes electrode projections; and the state-space stack models longer dependencies with linear sequence scaling. Alternating scan directions are appropriate for offline trial decoding because the complete adaptation or test trial is available at inference time. They would not, however, be causal in a real-time sample-by-sample deployment. A causal online version should use forward scans only and should be evaluated with latency measured on the intended hardware.

The adaptation losses also have distinct roles. Adversarial training changes features through a learned discriminator, whereas MK-MMD directly penalizes a kernel discrepancy. Their combination may be more robust than either mechanism alone, but marginal alignment can still bring different classes together. The temporal MMD term is deliberately assigned a smaller coefficient than final-feature MMD so that early temporal structure is regularized without dominating source classification.

Several limitations must be stated before interpreting the final results. First, BCI IV-2b has only three EEG channels and Zhou2016 has only four participants, so uncertainty across subjects can be large. Second, adaptation uses unlabeled data from a target user’s earlier sessions and is therefore not target-free domain generalization. Third, the model contains dataset-specific training budgets and trial windows; fair comparison requires matching the protocol of each cited baseline rather than comparing headline numbers from incompatible splits. Fourth, leakage control depends on routing only the target-adaptation loader, never the held-out test loader, to information maximization. Finally, repeated-seed runs, confidence intervals, and paired statistical tests are required before attributing a small improvement to the method

rather than to stochastic training.

Once the result tables are populated, the discussion should be extended with three evidence-based analyses: whether gains are consistent across subjects rather than driven by one fold; whether improvement remains after IM adaptation is removed; and whether the accuracy gain justifies additional parameters and adaptation time. These analyses are more informative than a single average accuracy and are necessary for a defensible comparison across the three datasets.

8. Conclusion

This paper formulates a cross-subject MI-EEG decoder that combines an EEG-specific multi-scale convolutional front-end, depthwise electrode filtering, alternating selective state-space sequence blocks, a local shortcut, and a grouped dilated TCN. The training objective integrates source classification with adversarial, trial-level MK-MMD, and temporal MK-MMD alignment, followed by constrained information-maximization adaptation of normalization affine parameters. A session-isolated LOSO protocol is specified for BCI Competition IV-2a, BCI Competition IV-2b, and Zhou2016 so that unlabeled target adaptation is separated from final target testing. Quantitative conclusions are deliberately deferred until the frozen clean protocol and statistical analysis have been completed. This separation between method design, target-data access, and final evaluation is intended to make subsequent claims reproducible and directly interpretable.

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